

BALANCED TWO-GENERATOR p -GROUPS OF DEFICIENCY ZERO AND UNBOUNDED NILPOTENCY CLASS

GROUP THEORY 50

ABSTRACT. For every prime $p > 3$ we construct an explicit family of finite two-generator p -groups of deficiency zero whose nilpotency classes tend to infinity. The construction gives, for every $n \geq 1$, a group of order p^{2n+1} and class $n + 1$. This gives an affirmative answer to Problem 17.113 of the Kourovka Notebook.

1. INTRODUCTION

The deficiency of a finite presentation

$$\langle x_1, \dots, x_d \mid r_1, \dots, r_s \rangle$$

is $d - s$, and the deficiency of a finitely presented group is the supremum of the deficiencies of its finite presentations. A presentation with the same number of generators and relators is usually called balanced. Balanced presentations of finite p -groups are constrained enough to be useful in topology and combinatorial group theory, but flexible enough to support groups of substantial nilpotency class.

Problem 17.113 of the Kourovka Notebook, proposed by J. Wiegold in the 17th issue, asks whether, for a fixed prime $p > 3$, there are two-generator finite p -groups of deficiency zero with arbitrarily high nilpotency class [1]. We give a uniform metacyclic construction.

2. THE FAMILY

Fix a prime $p > 3$ and an integer $n \geq 1$. Define

$$G_n = \langle a, b \mid b^{-1}ab = a^{1+p}, \quad b^{p^n} = a^{p^n} \rangle.$$

We use the commutator convention $[x, y] = x^{-1}y^{-1}xy$.

Theorem 2.1. *For every $n \geq 1$, the group G_n is a finite two-generator p -group of order p^{2n+1} , nilpotency class $n + 1$, and deficiency zero.*

Proof. Conjugating the relation $b^{p^n} = a^{p^n}$ by b gives

$$b^{p^n} = a^{p^n(1+p)}.$$

Comparison with the defining relation yields $a^{p^{n+1}} = 1$, and then also $b^{p^{n+1}} = 1$. Using $b^{-1}ab = a^{1+p}$, every word collects to $a^i b^j$. The relation $b^{p^n} = a^{p^n}$ allows us to choose

$$0 \leq i < p^{n+1}, \quad 0 \leq j < p^n,$$

so $|G_n| \leq p^{2n+1}$.

For the reverse inequality, put $q = 1 + p$ and let

$$A_n = \langle \alpha \rangle \cong C_{p^{n+1}}, \quad B_n = \langle \beta \rangle \cong C_{p^{n+1}}.$$

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Let B_n act on A_n by $\alpha^\beta = \alpha^q$. The standard lifting-the-exponent calculation gives

$$v_p(q^{p^k} - 1) = k + 1,$$

so this automorphism has order p^n . In $S_n = A_n \rtimes B_n$ the element

$$z = \beta^{p^n} \alpha^{-p^n}$$

is central of order p . Moreover $\langle z \rangle$ meets both A_n and B_n trivially. Hence

$$H_n = S_n / \langle z \rangle$$

has order p^{2n+1} . The images of α and β satisfy the defining relations of G_n , so there is an epimorphism $G_n \twoheadrightarrow H_n$. The upper bound already obtained forces $G_n \cong H_n$.

The first defining relation gives

$$[a, b] = a^p,$$

and therefore, for $k \geq 0$,

$$[a^{p^k}, b] = a^{p^{k+1}}.$$

It follows inductively that

$$\gamma_r(G_n) = \langle a^{p^{r-1}} \rangle \quad (2 \leq r \leq n+2).$$

Thus $\gamma_{n+1}(G_n) \neq 1$ and $\gamma_{n+2}(G_n) = 1$, so the class is $n+1$.

The displayed presentation has two generators and two relators, hence $\text{def}(G_n) \geq 0$. On the other hand, if a finite group has a presentation with d generators and s relators, abelianization and tensoring the relation matrix with $\mathbb{Q}$ show that $s \geq d$; thus every finite group has deficiency at most zero. Consequently $\text{def}(G_n) = 0$. $\square$

Corollary 2.2. *By Theorem 2.1, for every prime $p > 3$ there exist two-generator finite p -groups of deficiency zero and arbitrarily large nilpotency class.*

3. FURTHER REMARKS

The groups G_n are metacyclic central quotients of semidirect products of two cyclic p -groups. In particular, the growth of nilpotency class does not require a growing number of generators or relators. The same presentation also gives a concrete family suitable for computational experiments with balanced finite presentations.

The restriction $p > 3$ is the one appearing in the Kurovka problem. The construction itself uses only the familiar p -adic behavior of $1 + p$ for odd p ; we have not attempted here to optimize the small-prime range.

REFERENCES

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